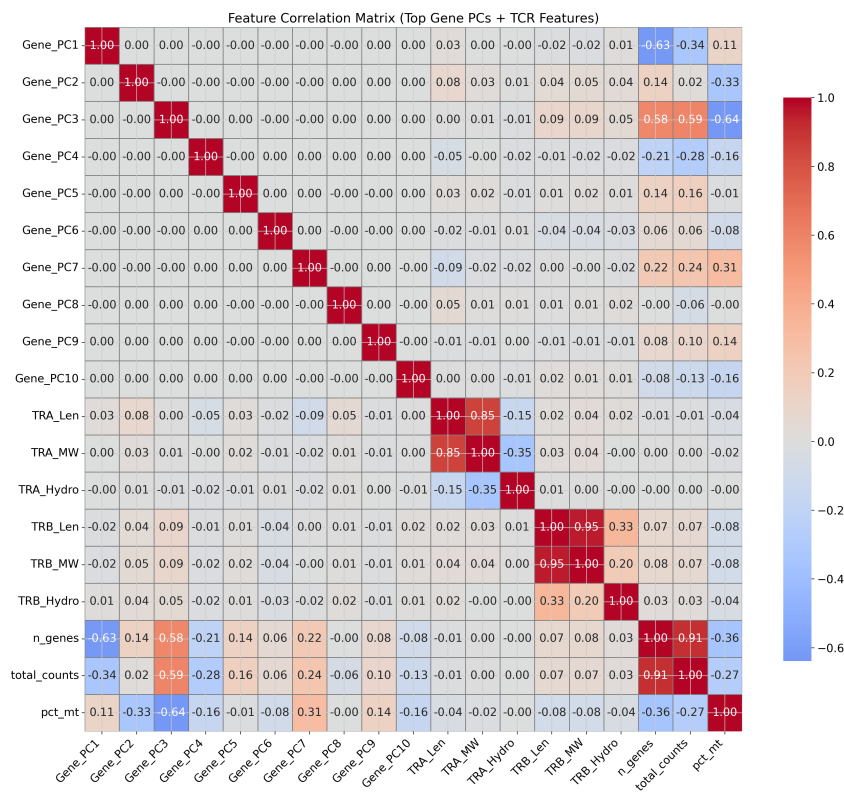
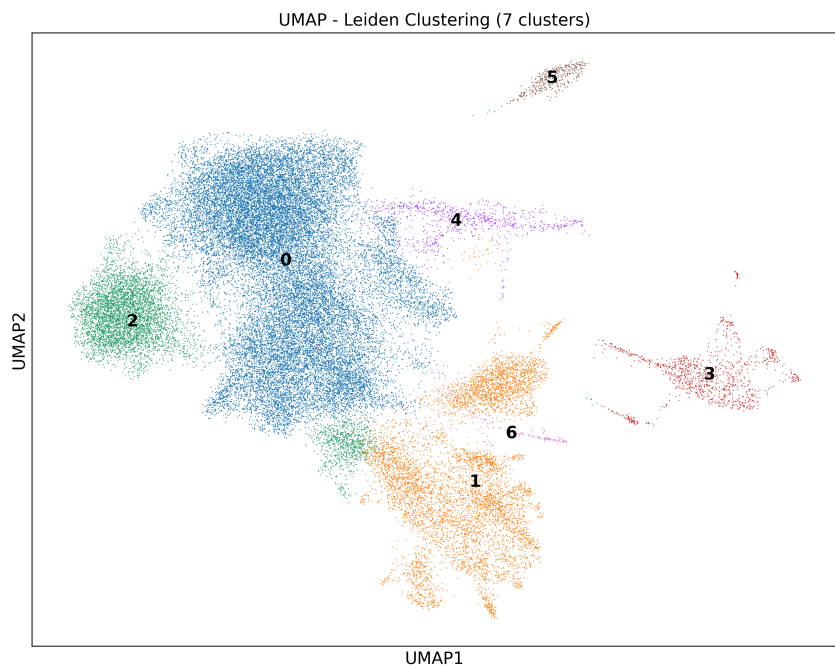




a: Pearson correlation matrix for the top engineered features (first 20 gene-expression PCs and select TCR/receptor summary statistics). Block-diagonal structure indicates orthogonality between transcriptomic and receptor feature families.



b: UMAP visualization of peripheral blood immune cells colored by Leiden cluster assignment (7 clusters, silhouette-optimized). Distinct clusters capture heterogeneous immune cell states relevant to treatment response.

Figure 1: Feature correlation structure and immune cell clustering. (a) The heatmap confirms low inter-family correlation, supporting the value of multimodal feature integration. (b) UMAP reveals well-separated immune subpopulations captured by the model.